Synchronous FIFO by using the UVM methodology-

Design Code

```
// Synchronous FIFO Design
module Sync_fifo #(parameter DATA_WIDTH=8, parameter
FIFO_DEPTH=16,parameter ADDR_WIDTH=4)(
        input logic clk,
        input logic rst_n,
        input logic rd_en,
        input logic wr_en,
        input logic [DATA_WIDTH-1:0] data_in,
        output logic [DATA_WIDTH-1:0] data_out,
      output logic underflow,
      output logic overflow,
      output logic full,
      output logic empty
);

logic [DATA_WIDTH-1:0] memory [0:FIFO_DEPTH-1];
logic [ADDR_WIDTH-1:0] wr_ptr, rd_ptr;
logic [ADDR_WIDTH:0] count;

always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
                wr_ptr   <= '0;
                rd_ptr   <= '0;
```

```
        count    <= '0;
        data_out <= '0;
        overflow <= 1'b0;
        underflow<= 1'b0;
    end
    else begin
            overflow  <= 1'b0;
            underflow <= 1'b0;

            if (wr_en && !full) begin
                    memory[wr_ptr] <= data_in;
                    wr_ptr <= (wr_ptr == FIFO_DEPTH-1) ? '0 : wr_ptr + 1'b1;
            end
            else if (wr_en && full) begin
                    overflow <= 1'b1;
            end

            if (rd_en && !empty) begin
                    data_out <= memory[rd_ptr];
                    rd_ptr <= (rd_ptr == FIFO_DEPTH-1) ? '0 : rd_ptr + 1'b1;
            end
            else if (rd_en && empty) begin
                    underflow <= 1'b1;
            end
```

```
            case ({(wr_en && !full), (rd_en && !empty)})
                2'b10: count <= count + 1'b1;
                2'b01: count <= count - 1'b1;
                2'b11: count <= count;
                default: count <= count;
            endcase
        end
end

always_comb begin
        full  = (count == FIFO_DEPTH);
        empty = (count == 0);
end
endmodule

//
interface fifo_if #(parameter DATA_WIDTH = 8,parameter FIFO_DEPTH =
16,parameter ADDR_WIDTH = 4)(input logic clk);
        logic               rst_n;
      logic                rd_en;
        logic               wr_en;
        logic [DATA_WIDTH-1:0] data_in;
        logic [DATA_WIDTH-1:0] data_out;
      logic               underflow;
      logic               overflow;
      logic                  full;
```

```
    logic               empty;

    clocking drv_cb @(posedge clk);
        default input #1step output #1ns;
        output rd_en, wr_en, data_in;
    endclocking

    clocking mon_cb @(posedge clk);
        default input #1step;
        input rd_en,wr_en,data_in,data_out,underflow,overflow,full,empty;
    endclocking

    modport DRV (clocking drv_cb,input rst_n);
   modport MON (clocking mon_cb,input rst_n);

endinterface

//
class seq_item extends uvm_sequence_item;
    rand bit      rd_en;
    rand bit      wr_en;
   rand bit [7:0] data_in;
    bit [7:0]     data_out;
    bit           underflow, overflow;
    bit           full, empty;
```

```
    function new(string name = "seq_item");
        super.new(name);
    endfunction

     `uvm_object_utils_begin(seq_item)
     `uvm_field_int(rd_en, UVM_ALL_ON)
    `uvm_field_int(wr_en, UVM_ALL_ON)
    `uvm_field_int(data_in,UVM_ALL_ON)
    `uvm_field_int(data_out,UVM_ALL_ON)
    `uvm_field_int(underflow,UVM_ALL_ON)
    `uvm_field_int(overflow, UVM_ALL_ON)
    `uvm_field_int(full, UVM_ALL_ON)
    `uvm_field_int(empty,UVM_ALL_ON)
     `uvm_object_utils_end

endclass

//
class base_seq extends uvm_sequence #(seq_item);
    seq_item req;
    `uvm_object_utils(base_seq)

    function new(string name = "base_seq");
        super.new(name);
```

```
        endfunction


        task body();
                `uvm_info(get_type_name(),"Base sequence has been
started",UVM_LOW)


                $display("[SEQ] : WRITE PHASE HAS BEEN STARTED");
                for (int i = 0; i < 16; i++) begin
                        `uvm_do_with(req, { wr_en == 1; rd_en == 0; data_in ==
(8'hA0 + i);})
                end
              $display("[SEQ] : OVER-WRITE PHASE HAS BEEN STARTED");
                repeat(5) begin
                        `uvm_do_with(req, { wr_en == 1; rd_en == 0;})
                end
              $display("[SEQ] : READ- PHASE HAS BEEN STARTED");
              repeat(21) begin
                   `uvm_do_with(req, { wr_en == 0; rd_en == 1;})
              end
/*              $display("[SEQ] : WRITE-READ PHASE HAS BEEN STARTED");
                repeat(3) begin
                        `uvm_do_with(req, { wr_en == 1; rd_en == 1;})
                end
              $display("[SEQ] : READ- PHASE HAS BEEN STARTED");
                repeat(3) begin
                        `uvm_do_with(req, { wr_en == 0; rd_en == 1;})
```

```
            end
            repeat(5) begin
                `uvm_do_with(req, { wr_en == 1; rd_en == 1;})
            end  */
          $display("[SEQ] : RANDOMIZE- PHASE HAS BEEN STARTED");
            repeat(10) begin
                `uvm_do(req)
            end
            `uvm_info(get_type_name(),"All the Base sequence have completely generated",UVM_LOW)
        endtask

endclass

//
class sequencer extends uvm_sequencer #(seq_item);
        `uvm_component_utils(sequencer)

        function new(string name = "sequencer", uvm_component parent = null);
                super.new(name, parent);
        endfunction

        function void build_phase(uvm_phase phase);
                super.build_phase(phase);
        endfunction
```

```
endclass

//
class driver extends uvm_driver #(seq_item);
	virtual fifo_if.DRV vif;
	`uvm_component_utils(driver)

	function new(string name = "driver", uvm_component parent = null);
		super.new(name, parent);
	endfunction

	function void build_phase(uvm_phase phase);
		super.build_phase(phase);

		if (!uvm_config_db#(virtual fifo_if.DRV)::get(this,"","vif",vif))
			`uvm_fatal(get_type_name(),"Virtual interface is not found
from top level")
	endfunction

	task run_phase(uvm_phase phase);
		wait(vif.rst_n == 1'b1);
		forever begin
			seq_item_port.get_next_item(req);
			@(vif.drv_cb);
			vif.drv_cb.wr_en   <= req.wr_en;
			vif.drv_cb.rd_en   <= req.rd_en;
```

```
                vif.drv_cb.data_in <= req.data_in;
                `uvm_info(get_type_name(),$sformatf("DRIVER:
WR_EN=%0d RD_EN=%0d
DATA_IN=%0h",req.wr_en,req.rd_en,req.data_in),UVM_LOW)
                seq_item_port.item_done();
            end
        endtask

endclass

//
class monitor extends uvm_monitor;
        virtual fifo_if vif;
        uvm_analysis_port #(seq_item) mon_ap;
        `uvm_component_utils(monitor)

        function new(string name = "monitor",uvm_component parent = null);
            super.new(name, parent);
            mon_ap = new("mon_ap", this);
        endfunction

        function void build_phase(uvm_phase phase);
            super.build_phase(phase);

            if (!uvm_config_db#(virtual fifo_if)::get(this,"","vif",vif))
```

```
                `uvm_fatal(get_type_name(),"Virtual interface is not found
from top level")
        endfunction


        task run_phase(uvm_phase phase);
                seq_item mon_item;
                forever begin
                        @(vif.mon_cb);
                        if (!vif.rst_n)
                                continue;
                        if (!vif.mon_cb.wr_en && !vif.mon_cb.rd_en)
                                continue;
                        mon_item = seq_item::type_id::create("mon_item");
                        mon_item.wr_en     = vif.mon_cb.wr_en;
                        mon_item.rd_en     = vif.mon_cb.rd_en;
                        mon_item.data_in   = vif.mon_cb.data_in;
                        #1;
                        mon_item.data_out  = vif.data_out;
                        mon_item.underflow = vif.underflow;
                        mon_item.overflow  = vif.overflow;
                        mon_item.full      = vif.full;
                        mon_item.empty     = vif.empty;
                        `uvm_info(get_type_name(),$sformatf("MONITOR:
WR_EN=%0d RD_EN=%0d DATA_IN=%0h DATA_OUT=%0h FULL=%0d
EMPTY=%0d OVERFLOW=%0d
UNDERFLOW=%0d",mon_item.wr_en,mon_item.rd_en,mon_item.data_in,mon_i
```

```
tem.data_out,mon_item.full,mon_item.empty,mon_item.overflow,mon_item.under
flow), UVM_LOW)


                mon_ap.write(mon_item);
        end
    endtask


endclass


//
class scoreboard extends uvm_scoreboard;
    `uvm_component_utils(scoreboard)
    uvm_analysis_imp #(seq_item, scoreboard) sb_imp;
    bit [7:0] ref_fifo[$];

    function new(string name = "scoreboard", uvm_component parent = null);
        super.new(name, parent);
        sb_imp = new("sb_imp", this);
    endfunction

    function void build_phase(uvm_phase phase);
        super.build_phase(phase);
        ref_fifo.delete();
    endfunction

    function void write(seq_item req);
```

```
        bit pre_full;
        bit pre_empty;
        bit write_accepted;
        bit read_accepted;
        bit expected_overflow;
        bit expected_underflow;
        bit expected_full;
        bit expected_empty;
        bit [7:0] expected_data;

        pre_full = (ref_fifo.size() == 16);
        pre_empty= (ref_fifo.size() == 0);

        write_accepted = req.wr_en && !pre_full;
        read_accepted  = req.rd_en && !pre_empty;

        expected_overflow = req.wr_en && pre_full;
      expected_underflow = req.rd_en && pre_empty;

        if (read_accepted) begin
                expected_data = ref_fifo.pop_front();

                if (req.data_out === expected_data) begin
                        `uvm_info("SCOREBOARD",$sformatf("READ PASS:
EXPECTED=%0h ACTUAL=%0h",expected_data,req.data_out),UVM_LOW)
                end
```

```
                else begin
                    `uvm_error("SCOREBOARD",$sformatf("READ FAIL: EXPECTED=%0h ACTUAL=%0h",expected_data,req.data_out))
                end
            end

            if (write_accepted) begin
                ref_fifo.push_back(req.data_in);
                `uvm_info("SCOREBOARD",$sformatf("WRITE PASS: DATA=%0h",req.data_in),UVM_LOW)
            end

            if (req.overflow === expected_overflow) begin
                `uvm_info("SCOREBOARD",$sformatf("OVERFLOW CHECK PASS: EXPECTED=%0d ACTUAL=%0d",expected_overflow,req.overflow),UVM_LOW)
            end
            else begin
                `uvm_error("SCOREBOARD",$sformatf("OVERFLOW CHECK FAIL: EXPECTED=%0d ACTUAL=%0d",expected_overflow,req.overflow))
            end

            if (req.underflow === expected_underflow) begin
                `uvm_info("SCOREBOARD",$sformatf("UNDERFLOW CHECK PASS: EXPECTED=%0d ACTUAL=%0d",expected_underflow,req.underflow),UVM_LOW)
            end
```

```
            else begin
                `uvm_error("SCOREBOARD",$sformatf("UNDERFLOW
CHECK FAIL: EXPECTED=%0d
ACTUAL=%0d",expected_underflow,req.underflow))
            end

            expected_full = (ref_fifo.size() == 16);
            expected_empty= (ref_fifo.size() == 0);

            if (req.full === expected_full) begin
                `uvm_info("SCOREBOARD","FULL CHECK
PASS",UVM_HIGH)
            end
            else begin
                `uvm_error("SCOREBOARD",$sformatf("FULL CHECK
FAIL: EXPECTED=%0d ACTUAL=%0d",expected_full,req.full))
            end

            if (req.empty === expected_empty) begin
                `uvm_info("SCOREBOARD","EMPTY CHECK
PASS",UVM_HIGH)
            end
            else begin
                `uvm_error("SCOREBOARD",$sformatf("EMPTY CHECK
FAIL: EXPECTED=%0d ACTUAL=%0d",expected_full,req.empty))
            end
        endfunction
```

```
endclass

//
class agent extends uvm_agent;
	`uvm_component_utils(agent)
	driver    drv;
	sequencer seqr;
	monitor   mon;

	function new(string name = "agent",uvm_component parent = null);
		super.new(name, parent);
	endfunction

	function void build_phase(uvm_phase phase);
		super.build_phase(phase);
		drv = driver::type_id::create("drv",this);
		seqr = sequencer::type_id::create("seqr",this);
		mon = monitor::type_id::create("mon",this);
	endfunction

	function void connect_phase(uvm_phase phase);
		super.connect_phase(phase);
		drv.seq_item_port.connect(seqr.seq_item_export);
	endfunction
```

```
endclass

//
class fifo_coverage extends uvm_subscriber #(seq_item);
        `uvm_component_utils(fifo_coverage)
        bit       cov_wr_en;
      bit       cov_rd_en;
      bit [7:0] cov_data_in;
        bit       cov_full;
      bit       cov_empty;
        bit       cov_overflow;
        bit       cov_underflow;

        covergroup fifo_cg;

        cp_operation:
      coverpoint {cov_wr_en, cov_rd_en} {
          bins idle        = {2'b00};
          bins write_only  = {2'b10};
          bins read_only   = {2'b01};
          bins read_write  = {2'b11};
      }

      cp_data:
```

```
coverpoint cov_data_in {
    bins zero     = {8'h00};
    bins max      = {8'hFF};
    bins low_data = {[8'h01:8'h3F]};
    bins mid_data = {[8'h40:8'hBF]};
      bins high_data = {[8'hC0:8'hFE]};
}

cp_full:
coverpoint cov_full {
    bins not_full = {0};
    bins full     = {1};
}

cp_empty:
coverpoint cov_empty {
    bins not_empty = {0};
    bins empty     = {1};
}

  cp_overflow:
coverpoint cov_overflow {
    bins no_overflow = {0};
    bins overflow    = {1};
}
```

```
    cp_underflow:
    coverpoint cov_underflow {
        bins no_underflow = {0};
        bins underflow    = {1};
    }

    cross_operation_status:
    cross cp_operation, cp_full, cp_empty;

    cross_operation_error:
    cross cp_operation, cp_overflow, cp_underflow;
    endgroup


function new(string name = "fifo_coverage", uvm_component parent = null);
          super.new(name, parent);
          fifo_cg = new();
endfunction

function void write(seq_item t);
          cov_wr_en    = t.wr_en;
        cov_rd_en    = t.rd_en;
        cov_data_in  = t.data_in;
        cov_full     = t.full;
```

```
        cov_empty    = t.empty;
        cov_overflow = t.overflow;
        cov_underflow = t.underflow;

      fifo_cg.sample();

   endfunction

endclass

//
class env extends uvm_env;
        `uvm_component_utils(env)
        agent          agt;
        scoreboard     sb;
        fifo_coverage  cov;

        function new(string name = "env",uvm_component parent = null);
                super.new(name, parent);
        endfunction

        function void build_phase(uvm_phase phase);
                super.build_phase(phase);
                agt = agent::type_id::create("agt",this);
                sb = scoreboard::type_id::create("sb",this);
```

```
        cov = fifo_coverage::type_id::create("cov",this);
    endfunction

    function void connect_phase(uvm_phase phase);
        super.connect_phase(phase);
        agt.mon.mon_ap.connect(sb.sb_imp);
        agt.mon.mon_ap.connect(cov.analysis_export);
    endfunction
endclass

//
class base_test extends uvm_test;
    `uvm_component_utils(base_test)
    env env_o;

    function new(string name = "base_test",uvm_component parent = null);
        super.new(name, parent);
    endfunction

    function void build_phase(uvm_phase phase);
        super.build_phase(phase);
        env_o = env::type_id::create("env_o",this);
    endfunction

    task run_phase(uvm_phase phase);
```

```
            base_seq bseq;
            phase.raise_objection(this);
            bseq = base_seq::type_id::create("bseq");
            bseq.start(env_o.agt.seqr);
            phase.drop_objection(this);
            `uvm_info(get_type_name(),"End of testcase",UVM_LOW)
        endtask
endclass

//
package fifo_pkg;

   `include "uvm_macros.svh"
   import uvm_pkg::*;
   `include "sequence_item.sv"
   `include "sequence.sv"
   `include "sequencer.sv"
   `include "driver.sv"
   `include "monitor.sv"
   `include "scoreboard.sv"
   `include "coverage.sv"
   `include "agent.sv"
   `include "environment.sv"
   `include "test.sv"
```

```
endpackage

//
`include "uvm_macros.svh"
import uvm_pkg::*;
import fifo_pkg::*;

module tb_top;
bit clk;
always #5 clk = ~clk;

fifo_if #(.DATA_WIDTH(8),.FIFO_DEPTH(16),.ADDR_WIDTH(4)) vif(clk);

Sync_fifo #(.DATA_WIDTH(8),.FIFO_DEPTH(16),.ADDR_WIDTH(4)) DUT
(.clk (vif.clk),.rst_n (vif.rst_n),.rd_en (vif.rd_en),.wr_en (vif.wr_en),.data_in
(vif.data_in),.data_out (vif.data_out),.underflow (vif.underflow),.overflow
(vif.overflow),.full (vif.full),.empty (vif.empty));

initial begin
        vif.rst_n   = 1'b0;
      vif.rd_en   = 1'b0;
      vif.wr_en   = 1'b0;
      vif.data_in = 8'h00;

        repeat(3)
        @(negedge clk);
```

```
    vif.rst_n = 1'b1;
end

initial begin
    uvm_config_db#(virtual fifo_if.DRV)::set(null, "*", "vif",vif);
    uvm_config_db#(virtual fifo_if)::set(null,"*","vif",vif);
end

initial begin
    run_test("base_test");
end

initial begin
    $fsdbDumpfile("sync_fifo.fsdb");
    $fsdbDumpvars(0,tb_top);
end
endmodule
```

File Content-

```
[b23n_sarif_727@prac3 sync_fifo_2]$ ls
agent.sv      environment.sv  monitor.sv      scoreboard.sv  simv
sync_fifo.fsdb  vc_hdrs.h
coverage.sv  fifo_if.sv      novas.conf      sequence_item.sv  simv.daidir
tb_top.sv      vdCov.conf
csrc         fifo_pkg.sv     novas_dump.log  sequencer.sv      simv.vdb          test.sv
vdCovLog
```

driver.sv inter.fsdb novas.rc sequence.sv sync_fifo_design.sv ucli.key verdiLog

Run Command:

[b23n_sarif_727@prac3 sync_fifo_2]$ vcs -sverilog sync_fifo_design.sv fifo_if.sv fifo_pkg.sv tb_top.sv -ntb_opts uvm -kdb -lca -debug_access+all -full64

[b23n_sarif_727@prac3 sync_fifo_2]$ ./simv

[b23n_sarif_727@prac3 sync_fifo_2]$ ./simv -verdi

[b23n_sarif_727@prac3 sync_fifo_2]$ verdi -cov -covdir simv.vdb/

Simulation Output-

UVM_INFO @ 0: reporter [RNTST] Running test base_test...

Verdi Loading libsscore_vcs202409.so

FSDB Dumper for VCS, Release Verdi_W-2024.09-1, Linux x86_64/64bit, 10/12/2024

(C) 1996 - 2024 by Synopsys, Inc.

Verdi FSDB WARNING: The FSDB file already exists. Overwriting the FSDB file may crash the programs that are using this file.

Verdi : Create FSDB file 'sync_fifo.fsdb'

Verdi : Begin traversing the scope (tb_top), layer (0).

Verdi : End of traversing.

UVM_INFO sequence.sv(11) @ 0: uvm_test_top.env_o.agt.seqr@@bseq [base_seq] Base sequence has been started

[SEQ] : WRITE PHASE HAS BEEN STARTED

UVM_INFO driver.sv(25) @ 35: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=a0

UVM_INFO driver.sv(25) @ 45: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=a1

UVM_INFO monitor.sv(37) @ 46: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=a0 DATA_OUT=0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 46: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=a0

UVM_INFO scoreboard.sv(54) @ 46: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 46: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 55: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=a2

UVM_INFO monitor.sv(37) @ 56: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=a1 DATA_OUT=0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 56: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=a1

UVM_INFO scoreboard.sv(54) @ 56: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 56: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 65: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=a3

UVM_INFO monitor.sv(37) @ 66: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=a2 DATA_OUT=0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 66: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=a2

UVM_INFO scoreboard.sv(54) @ 66: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 66: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 75: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=a4

UVM_INFO monitor.sv(37) @ 76: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=a3 DATA_OUT=0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 76: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=a3

UVM_INFO scoreboard.sv(54) @ 76: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 76: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 85: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=a5

UVM_INFO monitor.sv(37) @ 86: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=a4 DATA_OUT=0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 86: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=a4

UVM_INFO scoreboard.sv(54) @ 86: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 86: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 95: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=a6

UVM_INFO monitor.sv(37) @ 96: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=a5 DATA_OUT=0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 96: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=a5

UVM_INFO scoreboard.sv(54) @ 96: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 96: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 105: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=a7

UVM_INFO monitor.sv(37) @ 106: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=a6 DATA_OUT=0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 106: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=a6

UVM_INFO scoreboard.sv(54) @ 106: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 106: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 115: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=a8

UVM_INFO monitor.sv(37) @ 116: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=a7 DATA_OUT=0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 116: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=a7

UVM_INFO scoreboard.sv(54) @ 116: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 116: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 125: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=a9

UVM_INFO monitor.sv(37) @ 126: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=a8 DATA_OUT=0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 126: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=a8

UVM_INFO scoreboard.sv(54) @ 126: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 126: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 135: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=aa

UVM_INFO monitor.sv(37) @ 136: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=a9 DATA_OUT=0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 136: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=a9

UVM_INFO scoreboard.sv(54) @ 136: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 136: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 145: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=ab

UVM_INFO monitor.sv(37) @ 146: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=aa DATA_OUT=0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 146: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=aa

UVM_INFO scoreboard.sv(54) @ 146: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 146: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 155: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=ac

UVM_INFO monitor.sv(37) @ 156: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=ab DATA_OUT=0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 156: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=ab

UVM_INFO scoreboard.sv(54) @ 156: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 156: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 165: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=ad

UVM_INFO monitor.sv(37) @ 166: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=ac DATA_OUT=0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 166: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=ac

UVM_INFO scoreboard.sv(54) @ 166: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 166: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 175: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=ae

UVM_INFO monitor.sv(37) @ 176: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=ad DATA_OUT=0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 176: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=ad

UVM_INFO scoreboard.sv(54) @ 176: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 176: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 185: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=af

[SEQ] : OVER-WRITE PHASE HAS BEEN STARTED

UVM_INFO monitor.sv(37) @ 186: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=ae DATA_OUT=0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 186: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=ae

UVM_INFO scoreboard.sv(54) @ 186: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 186: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 195: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=b4

UVM_INFO monitor.sv(37) @ 196: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=af DATA_OUT=0 FULL=1 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 196: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=af

UVM_INFO scoreboard.sv(54) @ 196: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 196: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 205: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=e0

UVM_INFO monitor.sv(37) @ 206: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=b4 DATA_OUT=0 FULL=1 EMPTY=0 OVERFLOW=1 UNDERFLOW=0

UVM_INFO scoreboard.sv(54) @ 206: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=1 ACTUAL=1

UVM_INFO scoreboard.sv(61) @ 206: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 215: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=c2

UVM_INFO monitor.sv(37) @ 216: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=e0 DATA_OUT=0 FULL=1 EMPTY=0 OVERFLOW=1 UNDERFLOW=0

UVM_INFO scoreboard.sv(54) @ 216: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=1 ACTUAL=1

UVM_INFO scoreboard.sv(61) @ 216: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 225: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=69

UVM_INFO monitor.sv(37) @ 226: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=c2 DATA_OUT=0 FULL=1 EMPTY=0 OVERFLOW=1 UNDERFLOW=0

UVM_INFO scoreboard.sv(54) @ 226: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=1 ACTUAL=1

UVM_INFO scoreboard.sv(61) @ 226: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 235: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=1b

[SEQ] : READ- PHASE HAS BEEN STARTED

UVM_INFO monitor.sv(37) @ 236: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=69 DATA_OUT=0 FULL=1 EMPTY=0 OVERFLOW=1 UNDERFLOW=0

UVM_INFO scoreboard.sv(54) @ 236: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=1 ACTUAL=1

UVM_INFO scoreboard.sv(61) @ 236: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 245: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=27

UVM_INFO monitor.sv(37) @ 246: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=1b DATA_OUT=0 FULL=1 EMPTY=0 OVERFLOW=1 UNDERFLOW=0

UVM_INFO scoreboard.sv(54) @ 246: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=1 ACTUAL=1

UVM_INFO scoreboard.sv(61) @ 246: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 255: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=f0

UVM_INFO monitor.sv(37) @ 256: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=27 DATA_OUT=a0 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 256: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=a0 ACTUAL=a0

UVM_INFO scoreboard.sv(54) @ 256: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 256: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 265: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=97

UVM_INFO monitor.sv(37) @ 266: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=f0 DATA_OUT=a1 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 266: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=a1 ACTUAL=a1

UVM_INFO scoreboard.sv(54) @ 266: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 266: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 275: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=55

UVM_INFO monitor.sv(37) @ 276: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=97 DATA_OUT=a2 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 276: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=a2 ACTUAL=a2

UVM_INFO scoreboard.sv(54) @ 276: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 276: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 285: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=ef

UVM_INFO monitor.sv(37) @ 286: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=55 DATA_OUT=a3 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 286: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=a3 ACTUAL=a3

UVM_INFO scoreboard.sv(54) @ 286: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 286: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 295: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=c1

UVM_INFO monitor.sv(37) @ 296: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=ef DATA_OUT=a4 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 296: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=a4 ACTUAL=a4

UVM_INFO scoreboard.sv(54) @ 296: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 296: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 305: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=96

UVM_INFO monitor.sv(37) @ 306: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=c1 DATA_OUT=a5 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 306: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=a5 ACTUAL=a5

UVM_INFO scoreboard.sv(54) @ 306: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 306: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 315: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=fd

UVM_INFO monitor.sv(37) @ 316: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=96 DATA_OUT=a6 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 316: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=a6 ACTUAL=a6

UVM_INFO scoreboard.sv(54) @ 316: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 316: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 325: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=7f

UVM_INFO monitor.sv(37) @ 326: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=fd DATA_OUT=a7 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 326: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=a7 ACTUAL=a7

UVM_INFO scoreboard.sv(54) @ 326: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 326: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 335: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=e7

UVM_INFO monitor.sv(37) @ 336: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=7f DATA_OUT=a8 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 336: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=a8 ACTUAL=a8

UVM_INFO scoreboard.sv(54) @ 336: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 336: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 345: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=53

UVM_INFO monitor.sv(37) @ 346: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=e7 DATA_OUT=a9 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 346: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=a9 ACTUAL=a9

UVM_INFO scoreboard.sv(54) @ 346: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 346: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 355: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=31

UVM_INFO monitor.sv(37) @ 356: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=53 DATA_OUT=aa FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 356: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=aa ACTUAL=aa

UVM_INFO scoreboard.sv(54) @ 356: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 356: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 365: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=77

UVM_INFO monitor.sv(37) @ 366: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=31 DATA_OUT=ab FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 366: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=ab ACTUAL=ab

UVM_INFO scoreboard.sv(54) @ 366: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 366: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 375: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=d3

UVM_INFO monitor.sv(37) @ 376: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=77 DATA_OUT=ac FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 376: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=ac ACTUAL=ac

UVM_INFO scoreboard.sv(54) @ 376: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 376: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 385: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=9b

UVM_INFO monitor.sv(37) @ 386: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=d3 DATA_OUT=ad FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 386: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=ad ACTUAL=ad

UVM_INFO scoreboard.sv(54) @ 386: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 386: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 395: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=4c

UVM_INFO monitor.sv(37) @ 396: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=9b DATA_OUT=ae FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 396: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=ae ACTUAL=ae

UVM_INFO scoreboard.sv(54) @ 396: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 396: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 405: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=9a

UVM_INFO monitor.sv(37) @ 406: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=4c DATA_OUT=af FULL=0 EMPTY=1 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 406: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=af ACTUAL=af

UVM_INFO scoreboard.sv(54) @ 406: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 406: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 415: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=64

UVM_INFO monitor.sv(37) @ 416: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=9a DATA_OUT=af FULL=0 EMPTY=1 OVERFLOW=0 UNDERFLOW=1

UVM_INFO scoreboard.sv(54) @ 416: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 416: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=1 ACTUAL=1

UVM_INFO driver.sv(25) @ 425: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=47

UVM_INFO monitor.sv(37) @ 426: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=64 DATA_OUT=af FULL=0 EMPTY=1 OVERFLOW=0 UNDERFLOW=1

UVM_INFO scoreboard.sv(54) @ 426: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 426: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=1 ACTUAL=1

UVM_INFO driver.sv(25) @ 435: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=2f

UVM_INFO monitor.sv(37) @ 436: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=47 DATA_OUT=af FULL=0 EMPTY=1 OVERFLOW=0 UNDERFLOW=1

UVM_INFO scoreboard.sv(54) @ 436: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 436: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=1 ACTUAL=1

UVM_INFO driver.sv(25) @ 445: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=68

[SEQ] : RANDOMIZE- PHASE HAS BEEN STARTED

UVM_INFO monitor.sv(37) @ 446: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=2f DATA_OUT=af FULL=0 EMPTY=1 OVERFLOW=0 UNDERFLOW=1

UVM_INFO scoreboard.sv(54) @ 446: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 446: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=1 ACTUAL=1

UVM_INFO driver.sv(25) @ 455: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=1 DATA_IN=5a

UVM_INFO monitor.sv(37) @ 456: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=68 DATA_OUT=af FULL=0 EMPTY=1 OVERFLOW=0 UNDERFLOW=1

UVM_INFO scoreboard.sv(54) @ 456: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 456: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=1 ACTUAL=1

UVM_INFO driver.sv(25) @ 465: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=0 DATA_IN=6d

UVM_INFO monitor.sv(37) @ 466: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=1 DATA_IN=5a DATA_OUT=af FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=1

UVM_INFO scoreboard.sv(50) @ 466: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=5a

UVM_INFO scoreboard.sv(54) @ 466: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 466: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=1 ACTUAL=1

UVM_INFO driver.sv(25) @ 475: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=b1

UVM_INFO driver.sv(25) @ 485: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=61

UVM_INFO monitor.sv(37) @ 486: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=0 RD_EN=1 DATA_IN=b1 DATA_OUT=5a FULL=0 EMPTY=1 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 486: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=5a ACTUAL=5a

UVM_INFO scoreboard.sv(54) @ 486: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 486: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 495: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=0 DATA_IN=93

UVM_INFO monitor.sv(37) @ 496: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=61 DATA_OUT=5a FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 496: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=61

UVM_INFO scoreboard.sv(54) @ 496: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 496: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 505: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=1 DATA_IN=7f

UVM_INFO driver.sv(25) @ 515: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=0 DATA_IN=94

UVM_INFO monitor.sv(37) @ 516: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=1 DATA_IN=7f DATA_OUT=61 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 516: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=61 ACTUAL=61

UVM_INFO scoreboard.sv(50) @ 516: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=7f

UVM_INFO scoreboard.sv(54) @ 516: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 516: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 525: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=1 RD_EN=1 DATA_IN=e4

UVM_INFO monitor.sv(37) @ 526: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=0 DATA_IN=94 DATA_OUT=61 FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(50) @ 526: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=94

UVM_INFO scoreboard.sv(54) @ 526: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 526: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 535: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=1 DATA_IN=ba

UVM_INFO monitor.sv(37) @ 536: uvm_test_top.env_o.agt.mon [monitor] MONITOR: WR_EN=1 RD_EN=1 DATA_IN=e4 DATA_OUT=7f FULL=0 EMPTY=0 OVERFLOW=0 UNDERFLOW=0

UVM_INFO scoreboard.sv(41) @ 536: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: EXPECTED=7f ACTUAL=7f

UVM_INFO scoreboard.sv(50) @ 536: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: DATA=e4

UVM_INFO scoreboard.sv(54) @ 536: uvm_test_top.env_o.sb [SCOREBOARD] OVERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO scoreboard.sv(61) @ 536: uvm_test_top.env_o.sb [SCOREBOARD] UNDERFLOW CHECK PASS: EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(25) @ 545: uvm_test_top.env_o.agt.drv [driver] DRIVER: WR_EN=0 RD_EN=0 DATA_IN=3e

UVM_INFO sequence.sv(40) @ 545: uvm_test_top.env_o.agt.seqr@@bseq [base_seq] All the Base sequence have completely generated

UVM_INFO test.sv(21) @ 545: uvm_test_top [base_test] End of testcase

UVM_INFO /usr/synopsys/vcs/W-2024.09-1/etc/uvm/base/uvm_objection.svh(1274) @ 545: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase

--- UVM Report Summary ---

** Report counts by severity

UVM_INFO : 241

UVM_WARNING : 0

UVM_ERROR : 0

UVM_FATAL : 0

** Report counts by id

[RNTST] 1

[SCOREBOARD] 136

[TEST_DONE] 1

[base_seq] 2

[base_test] 1

[driver] 52

[monitor] 48

$finish called from file "/usr/synopsys/vcs/W-2024.09-1/etc/uvm/base/uvm_root.svh", line 437.

$finish at simulation time 545

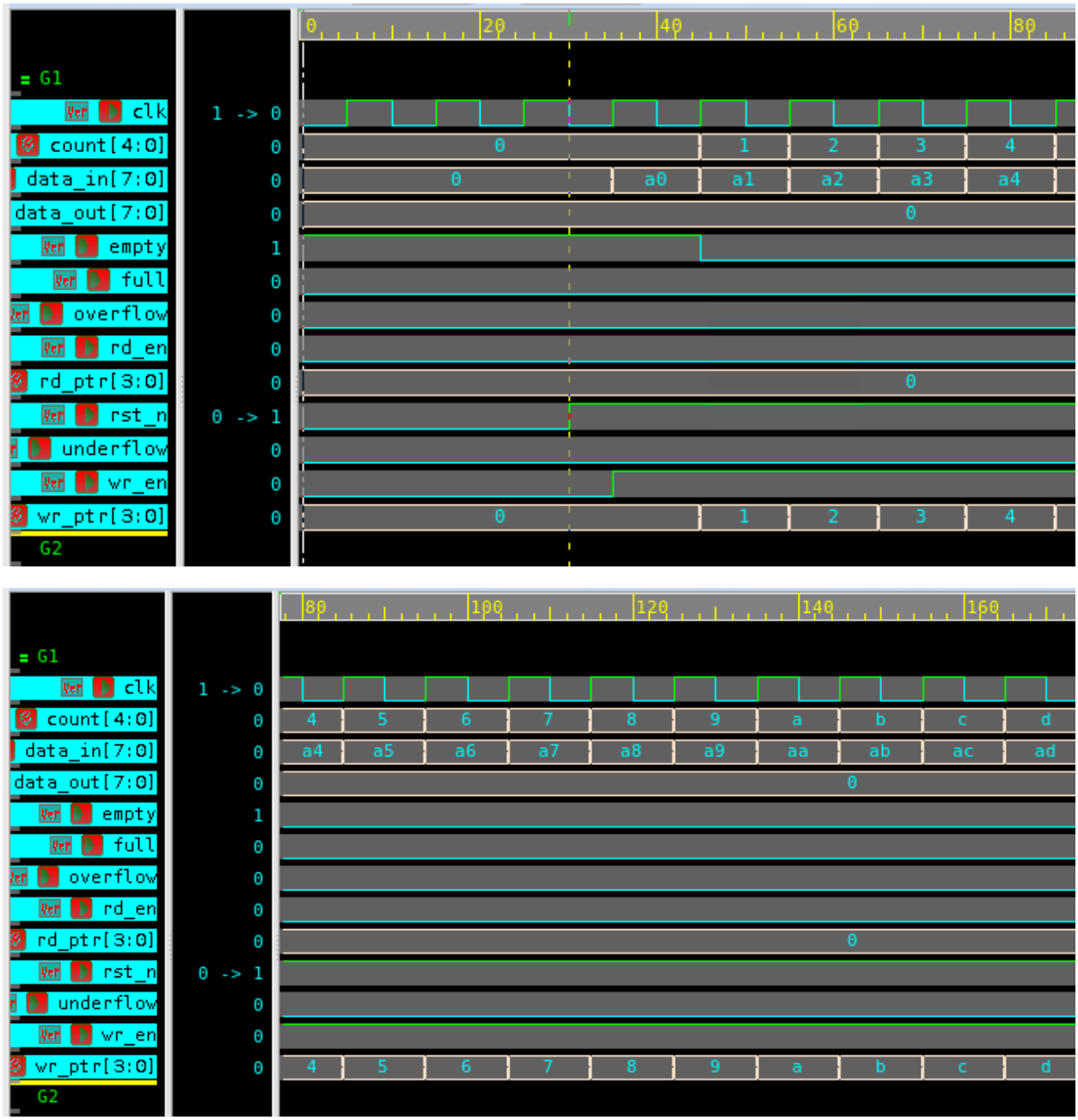
V C S S i m u l a t i o n R e p o r t

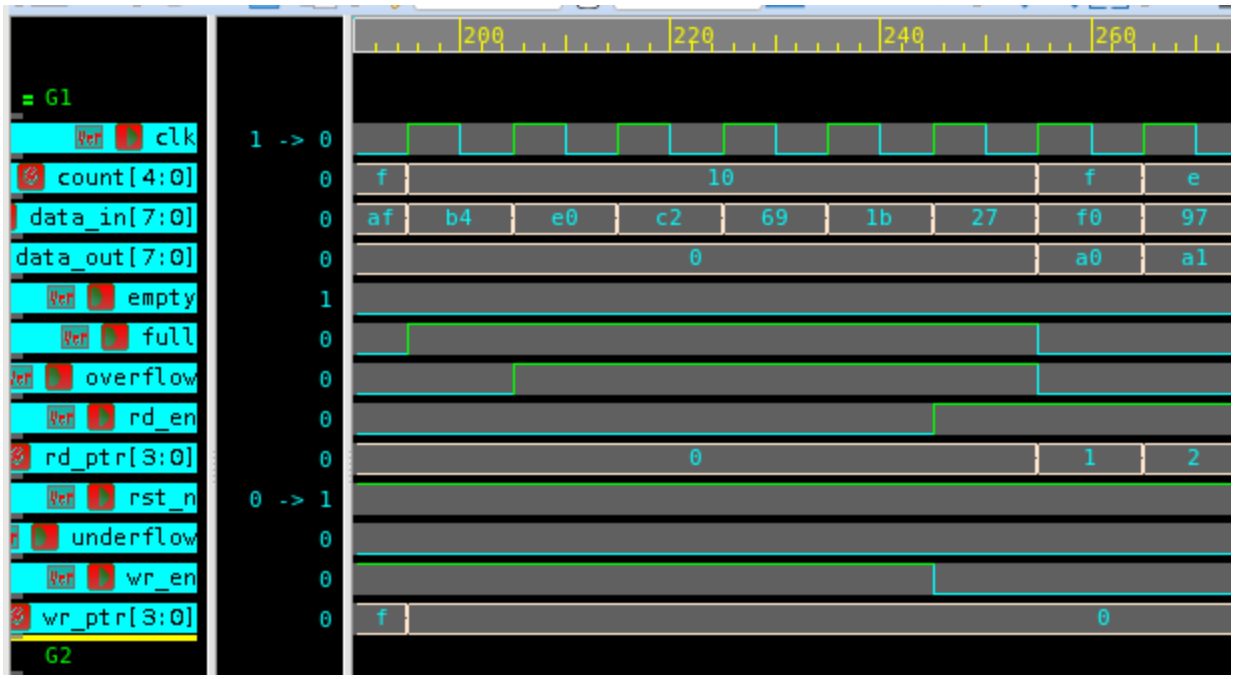
Time: 545

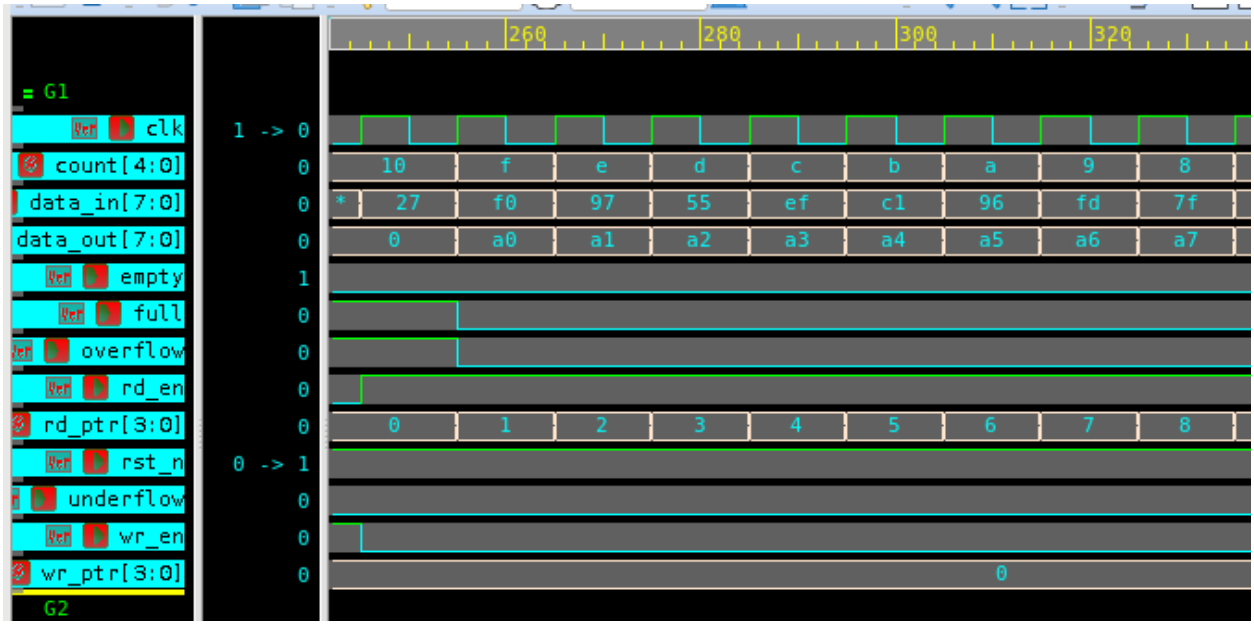
CPU Time: 0.290 seconds; Data structure size: 0.6Mb

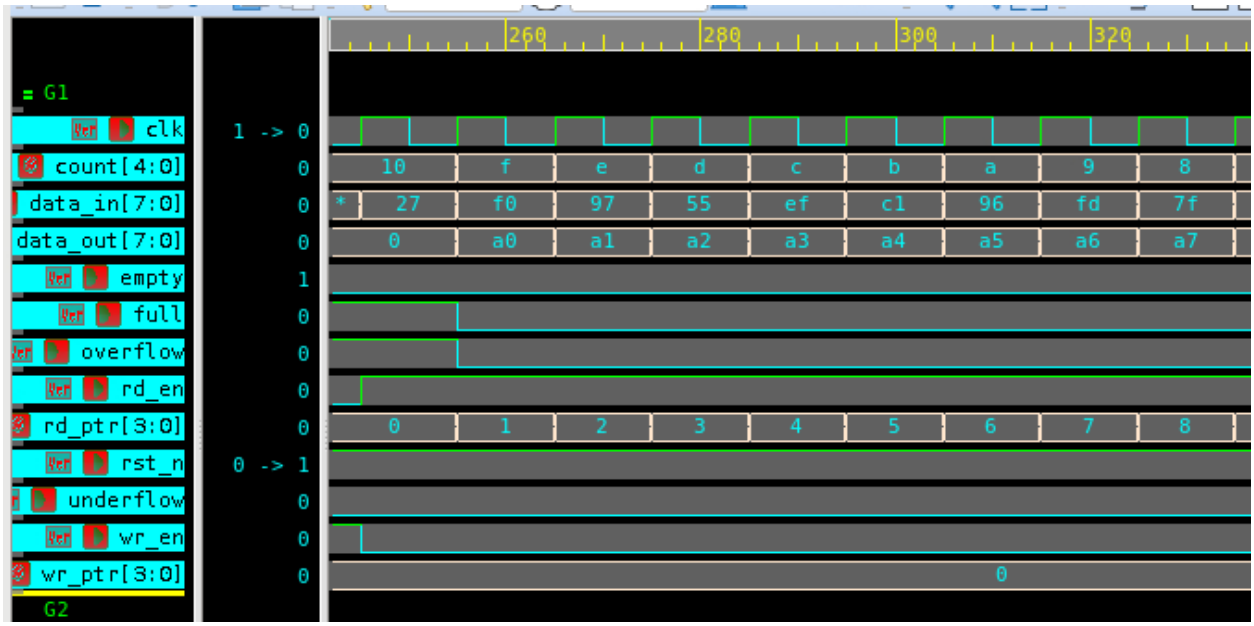
Sun Sep 13 07:27:36 2026

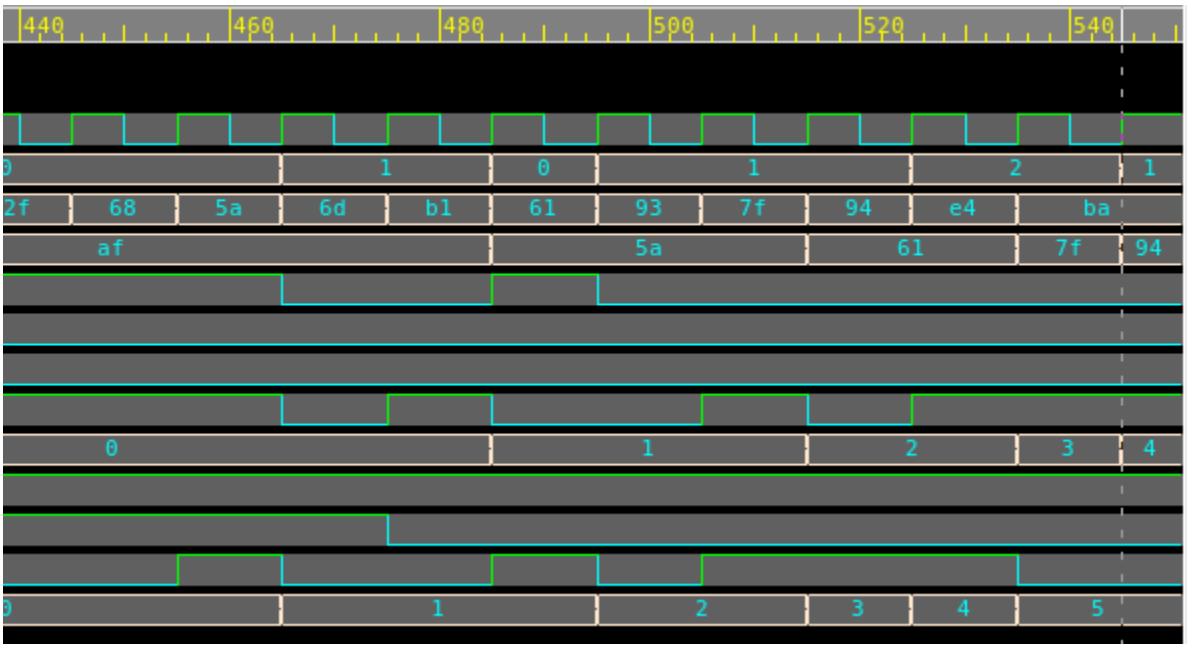
Waveform Analysis-

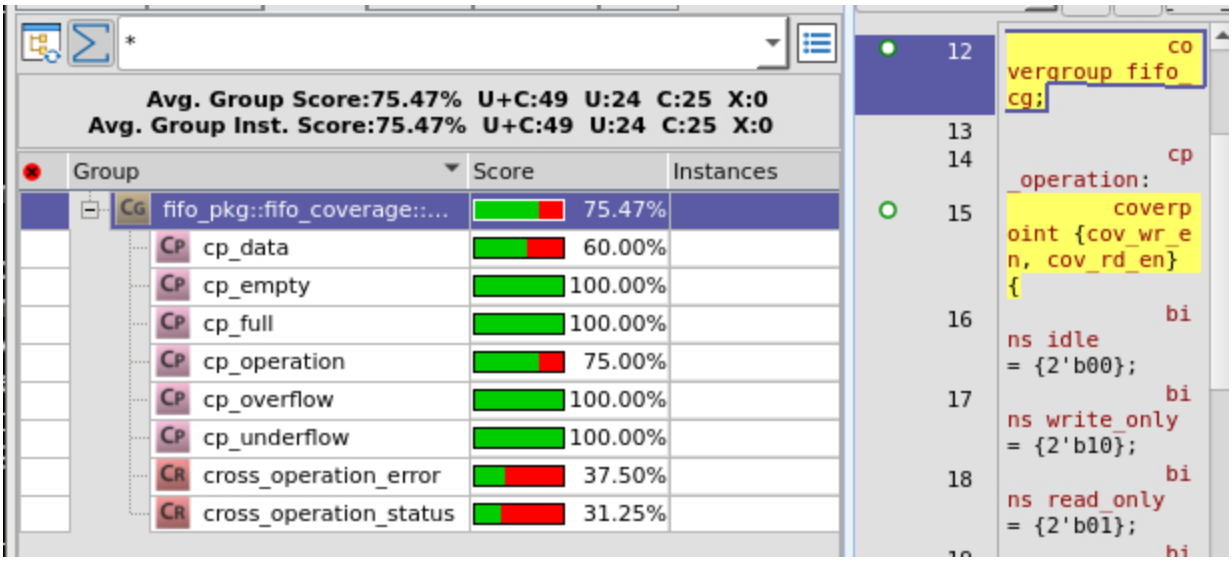



200
220
240
260
G1
clk
count[4:0]
data_in[7:0]
data_out[7:0]
empty
full
overflow
rd_en
rd_ptr[3:0]
rst_n
underflow
wr_en
wr_ptr[3:0]
G2
1 -> 0
0 -> 1
f
10
e
af
b4
e0
c2
69
1b
27
f0
97
0
a0
a1
1
2


260
280
300
320
G1
clk
count[4:0]
data_in[7:0]
data_out[7:0]
empty
full
overflow
rd_en
rd_ptr[3:0]
rst_n
underflow
wr_en
wr_ptr[3:0]
G2
1 -> 0
0 -> 1
10
f
e
d
c
b
a
9
8
27
f0
97
55
ef
c1
96
fd
7f
0
a0
a1
a2
a3
a4
a5
a6
a7
1
2
3
4
5
6
7


260
280
300
320
G1
clk
count[4:0]
data_in[7:0]
data_out[7:0]
empty
full
overflow
rd_en
rd_ptr[3:0]
rst_n
underflow
wr_en
wr_ptr[3:0]
G2
1 -> 0
0 -> 1
10 f e d c b a 9 8
27 f0 97 55 ef c1 96 fd 7f
0 a0 a1 a2 a3 a4 a5 a6 a7
0 1 2 3 4 5 6 7 8


440
460
480
500
520
540
0 1 0 1 2 1
2f 68 5a 6d b1 61 93 7f 94 e4 ba
af 5a 61 7f 94
0 1 2 3 4
0 1 2 3 4 5


*
Avg. Group Score:75.47% U+C:49 U:24 C:25 X:0
Avg. Group Inst. Score:75.47% U+C:49 U:24 C:25 X:0
Group
Score
Instances
fifo_pkg::fifo_coverage::...
75.47%
cp_data
60.00%
cp_empty
100.00%
cp_full
100.00%
cp_operation
75.00%
cp_overflow
100.00%
cp_underflow
100.00%
cross_operation_error
37.50%
cross_operation_status
31.25%
12
covergroup fifo_cg;
13
14
cp_operation:
15
coverpoint {cov_wr_en, cov_rd_en} {
16
bins idle = {2'b00};
17
bins write_only = {2'b10};
18
bins read_only = {2'b01};